

THE BIVARIATE GAMMA CORRELATION SURFACE

by

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SUMMARY

Pearson (1900) derived the tetrachoric series for the bivariate normal surface. Gonin (1966) derived similar series for the Poisson and binomial frequency surfaces. In the present note, we derive a similar series for the bivariate gamma correlation surface. This series is useful for the estimation of the correlation coefficient in 2x2 tables derived from a bivariate gamma distribution.

1. The Bivariate Gamma Distribution

Kibble (1941) derived the bivariate gamma frequency function $g_a(x, y; \rho)$ as a series bilinear in Laguerre polynomials in x and y :

$$(1) \quad g_a(x, y; \rho) = g_a(x)g_a(y) \left[1 + \sum_{r=1}^{\infty} \frac{r! \Gamma(a)}{\Gamma(a+r)} \rho^r L_r^{(a-1)}(x) L_r^{(a-1)}(y) \right],$$

where $0 \leq x, 0 \leq y$ and $g_a(x) = e^{-x} x^{a-1} / \Gamma(a)$. The set of Laguerre polynomials $L_r^{(a-1)}(x)$ is defined by [Szegő (1959)]

$$(2) \quad L_0^{(a-1)}(x) = 1, L_r^{(a-1)}(x) g_a(x) = \left(\frac{-d}{dx} \right)^r [x^r g_a(x) / r!] \quad \text{for } r=1, 2, \dots$$

Formula (1) is a particular form of the general canonical expansion of bivariate distributions given by Lancaster (1958, 1963).

2. The Bivariate Gamma Tetrachoric Correlation

Consider a sample of size N derived from the bivariate gamma distribution (1), and let the sample be grouped in the form of a fourfold table with frequencies a, b, c, d in the usual form ($a+b+c+d=N$). Let h and k be the points of dichotomy on the marginals, i.e.

$$(3) \quad \frac{b+d}{N} = \int_h^{\infty} g_a(x) dx, \quad \frac{c+d}{N} = \int_k^{\infty} g_a(y) dy.$$

Now, formula (2) may be put in the form

$$(4) \quad L_r^{(a-1)}(x) g_a(x) = \frac{\Gamma(a+r)}{\Gamma(a)r!} \left(\frac{-d}{dx} \right)^r [g_{a+r}(x)].$$

Using (4), we have

$$\begin{aligned}
 (5) \quad \int_h^\infty L_r^{(a-1)}(x) g_a(x) dx &= \frac{\Gamma(a+r)}{\Gamma(a) r!} \left[\left(\frac{-d}{dx} \right)^{r-1} g_{a+r}(x) \right]_\infty^h \\
 &= \frac{\Gamma(a+r)}{\Gamma(a) r!} \cdot \frac{\Gamma(a+1)(r-1)!}{\Gamma(a+r)} \left[L_{r-1}^{(a)}(x) g_{a+1}(x) \right]_\infty^h \\
 &= \frac{a}{r} L_{r-1}^{(a)}(h) g_{a+1}(h) = \frac{h}{r} L_{r-1}^{(a)}(h) g_a(h).
 \end{aligned}$$

Equating the observed proportion d/N to its expected value, and using (1), (3) and (5) we have

$$\begin{aligned}
 \frac{d}{N} &= \int_h^\infty \int_k^\infty g_a(x, y; \rho) dx dy \\
 &= \frac{(b+d)(c+d)}{N^2} + \sum_{r=1}^\infty \frac{r! \Gamma(a)}{\Gamma(a+r)} \rho^r \int_h^\infty L_r^{(a-1)}(x) g_a(x) dx \int_k^\infty L_r^{(a-1)}(y) g_a(y) dy
 \end{aligned}$$

and hence

$$(6) \quad \frac{ad-bc}{N^2} = h k g_a(h) g_a(k) \sum_{r=1}^\infty \frac{r! \Gamma(a) \rho^r}{r^2 \Gamma(a+r)} L_{r-1}^{(a)}(h) L_{r-1}^{(a)}(k).$$

If we normalize the Laguerre polynomials by taking

$$L_{r-1}^{*(a)}(x) = L_{r-1}^{(a)}(x) \left[\frac{\Gamma(a+r)}{\Gamma(a+1)(r-1)!} \right]^{-\frac{1}{2}},$$

We may write (6) in the form

$$(7) \quad \frac{a(ad-bc)}{N^2 h k g_a(h) g_a(k)} = \sum_{r=1}^\infty \frac{\rho^r}{r} L_{r-1}^{*(a)}(h) L_{r-1}^{*(a)}(k).$$

It is interesting to note that the right hand side of (7) has the same form as Pearson's (1900) tetrachoric series (with the normalized Laguerre polynomials replacing the normalized Hermite-Chebyshev polynomials). However, the left hand side of (7) contains the additional factor $a(hk)^{-1}$, as compared with Pearson's equation [formula (13) of Hamdan (1968)].

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